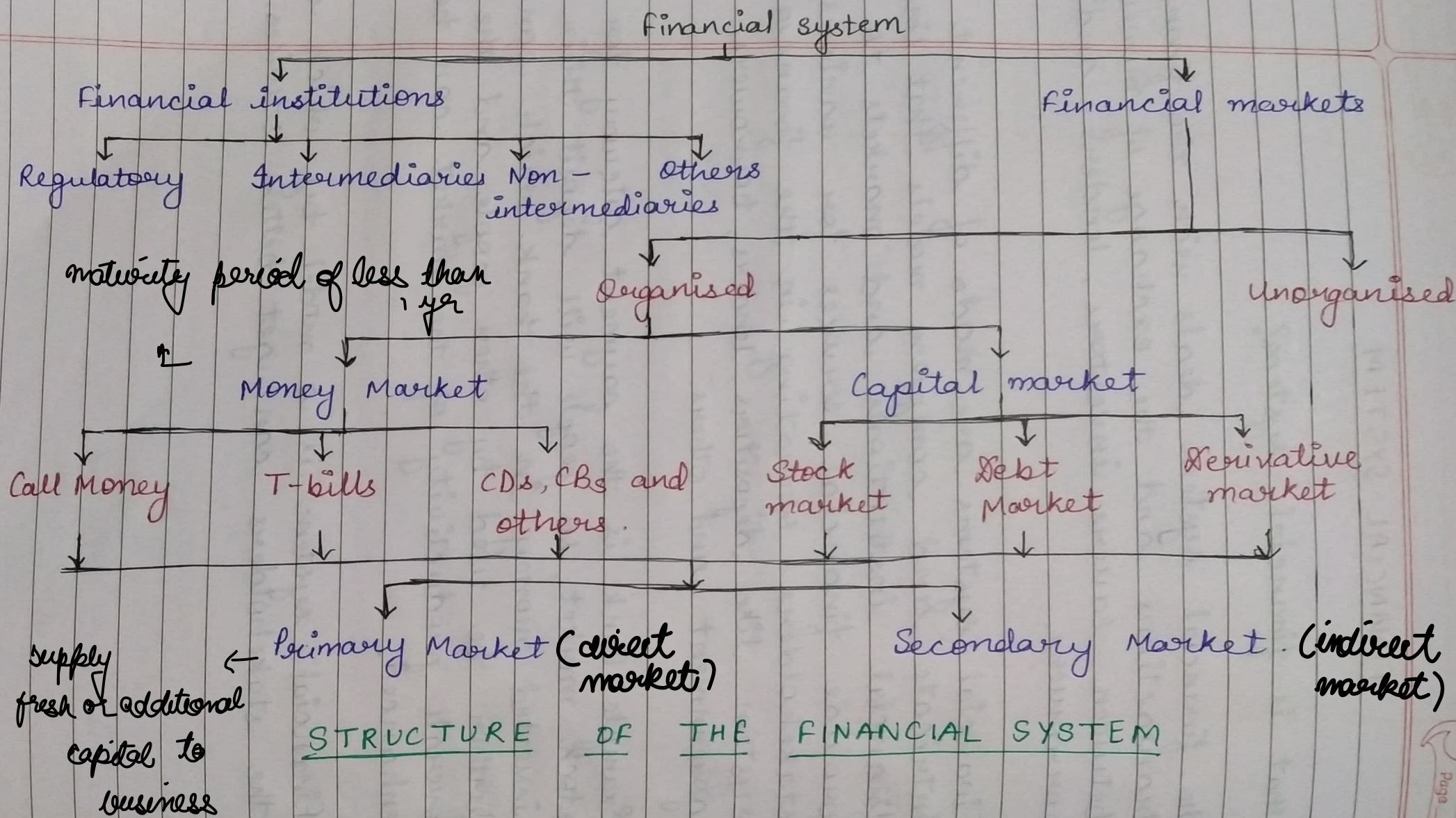


FINANCIAL SYSTEM

What is financial system?

- The financial system deals with the financial transactions and the exchange of money between savers, investors, lenders and borrowers.
- Financial systems are made of different intricate and complex models that link financial institutions and markets to provide financial services for various stakeholders operating in the financial system like depositors, lenders, borrowers, government and others.
- Example: Bank is the payment gateway for the stock market, so people will deposit their invested money in the bank which can further be used by other people and the overall productivity of the market can be enhanced.
- Financial system is a model by which all the stakeholders can get better services.



- Regulatory bodies:

Several bodies set up the regulatory framework of the Indian financial system. They are all there to ensure parity and responsibility among participants in that particular sub-sector. Every regulator is instrumental in making sure that the interests of the investors and all other parties are not compromised and that there is fairness in the financial system of India.

Financial regulators in India are:

- 1) **SEBI:** The market regulator in the Indian Capital market is the Securities and Exchange Board of India (SEBI).
- 2) **IRDA:** The Insurance Regulatory and Development Authority (IRDA) does the same for the insurance sector.
- 3) **RBI:** Reserve Bank of India (RBI) conducts the country's monetary policy.
- 4) **PFRDA:** Pension Funds Regulatory and Development Authority (PFRDA) regulates pensions.
- 5) **MCA:** Ministry of Corporate Affairs (MCA) regulates the corporate sector.

- Money market deals with short term securities and capital market deals with long term securities.

- CDS → Certificate of deposits
CBS → Commercial borrowings.

* Classification of Financial Institutions

- Banking and Non-Banking
- Banks provide transaction services.
- Create deposits or credits.
- Subject to legal reserve requirements.
- Can advance credit by creating claims against themselves.
- Other institutions can lend ~~to~~ only out of resources put at their disposal by the savers.
- Examples of non-banking financial institutions are Life Insurance Corporation (LIC), Mutual Fund Institutions (MFI), and other Non-Banking Financial Companies (NBFCs).
- According to savers banks are "creators" of credit, and non-banking institutions are "purveyors" of credit.
- Intermediaries vs. Non-intermediaries
- Intermediaries intermediate between savers and investors.
Savers deposit their money and are paid

interest by the banks, $\therefore$ Savings are an asset for the savers and a liability for the banks. The bank gives out loan from the savings, on which the borrower pays interest to the bank, $\therefore$ loans are an asset for the bank and liability for the people (borrowers).

- They lend money as well as mobilise savings.
- Their liabilities are towards the ultimate savers, while their assets are from the investors or ~~bank~~ borrowers.
- All banking institutions are intermediaries and LIC & GIC are some of the non-banking financial intermediaries (NBFI).
- Non-intermediary institutions do the loan business but their resources are not directly obtained from the savers. Example: IFC, NABARD.
International Financial Corporation (IFC)
National Bank for Agriculture and Rural Development (NABARD).
- Non-intermediaries do not deal with the savers, it only deals (gives loans) to the investors or the other sectors e.g. farmers.
- Non-intermediaries exist in the market for the development of some particular

sectors.

Example: NABARD for agriculture sector.

IDBI → Industrial Development Bank of India (IDBI). for the industrial sector.
Over the time IDBI has converted to a bank.

HDFC Ltd. → Parent Company → Housing Loans.

* Classification of Financial Markets

• Money and Capital Markets

- This conventional distinction is based on the differences in the period of maturity of financial assets issued in these markets.
- While the money market deals in the short-term claims (with a period of maturity of 1 year or less), the capital markets does so in the long-term (maturity period above 1 year) claims.

• Primary and Secondary Markets

- Primary markets deal in the new financial claims or new securities
- Secondary markets deal in securities already issued or existing or outstanding.
- Example: IPO: Initial Public Offerings (IPO)

Date _____
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is the market where a company issues its shares first time to the public to raise money. These are the new shares that are added in the existing market. This is the primary market.

- Once the shares have been traded in IPO and listed in the stock exchange then it becomes an asset of the secondary market.
- If the secondary market gain a profit of some amount then it is due to the fluctuation of demand and supply and not due to addition of new shares in the market.
- When the demand rises & becomes greater than supply in the secondary market then it gains positive profit.
- The development of secondary market is linked to demand & supply of the particular shares in a particular time period. The value of the market depends upon the demand & supply forces.
- In IPO market, demand & supply has not been created yet, but when the shares are traded in that market then they come to the secondary market.

- ∴ The value of the shares in the market are based on the demand of the shares. It is not on the basis of addition to the existing shares but due to price appreciation on the basis of ~~availability~~ demand & supply of the shares.

sub

• Equilibrium in Financial System

- Demand ^{of money} comes from investment side.
- Supply ^{of money} comes from saving side.
- If Savings = Investment then the money market is in eq.

- Equilibrium is established when the expected demand for funds (credit) for short-term & long-term investment matches with the planned supply of funds generated out of savings and credit creation.

- Biggest saver is the household sector.

• Determinants of supply of Funds

Factors affecting supply of funds

- Aggregate savings by the household sector, business sector and the government sector.
- ① Aggregate savings → household, corporate, govt.
- Business sector → Corporate profit deposited as savings.

② • Level of current & expected income.

More income $\Rightarrow$ more savings.

- Cyclical changes in income, age wise variations in income, distribution of income in the economy, degree of certainty of income.
- In case of distribution of income, ~~the~~ the MPC for rich is less & MPS is more and on the other hand, MPC for poor is more & MPS is less.
- Rich have savings in the bank for which they are paid interest & they become richer day-by-day and on the other hand poor get poorer day-by-day.
- $\therefore$ If income of rich is very & that of poor is very low then consumption will go down & aggregate demand will decrease, $\therefore$ it will lead to less production & total income generated in the country will decrease.
- Therefore, in this case, even if there are savings then they cannot be utilized for investment as there is no demand for consumer goods.
- Wealth: Total assets that any individual or institution have. e.g. Real estate, etc.

- Inflation
- Desire to provide for old age, family members, contingencies
- Rate of interest (loans & deposits)
~~Rate of interest is~~
- Supply of funds is directly proportional to rate of interest as r is the reward for savings $\therefore$ as r increases, savings increase
- Availability of savings media with preferred investments characteristics.
- But lower interest rate leads to more investment & more generation of income.
- ~~Even~~ Due to lack of resources & information, some people willing to save money are not able to save money.
- There are different options available to save money & people may have their preferred investment characteristics.
- Development of banks & other financial institutions.
- More developed banking & financial sector can lead to higher supply of money & higher investments.

* Determinants of Demands for Funds

circulating

- Investment in fixed and (working) capital

Money is required to fulfill both, the short term investments or requirements & the short term obligations.

- The current level of capital stock.
The companies demand funds on the basis of the availability of capital stock in the company.

Capital stock is the amount of common & preferred shares that a company is authorized to issue, according to its corporate charter. Capital stock can only be issued by the company and is the maximum number of shares that can ever be outstanding.

- Capacity utilisation

Demand of funds depends on how much capacity the company has already utilised & how much capacity is left to ~~at~~ utilise in order to maximize the revenue.

~~Capit~~ Capacity utilisation can be in terms of the fixed ^{asset} ~~capital~~ & the variable asset. Because if there is some resource that is still unutilized then the extra amount or

funding is not required..

The desired capital stock, which is influenced by business expectations (prospects) regarding future demand for goods (sales), prices, Government policies, and profitability.

The desired capital stock is the amount of capital stock desired by the company which it can fully & efficiently utilise.

Business expectations (prospects) → Opportunities in the market.

Demand for goods (sales) → Companies will only produce the product when it is demanded in the market.

Price → Depends on the market structure:
Price in perfect competition is less than price in monopoly. Depending upon the number of competitors in the market, price of the product is decided.

Government policies → Monetary & fiscal policies.

- Availability of internal funds. → Retained earnings
= Total profit - Interest on the loan. - Tax

Interest payments are tax deductible in nature.

The fund available with the company after the payment of interest, tax and dividends are the retained earnings.

If the availability of internal funds is higher then the company may not need much extra funds for investment.

- Cost of funds. → Interest rate is a major factor for raising the funds.

A company can raise capital from two sources: debt and equity.

Debt can be borrowed from banks by the company. Or company's can issue bonds to raise capital from the particular market.

Or the company can raise capital by issuing equities to its shareholders. This is called external equity.

- Technological changes.
To utilise the better & advanced technology, more funds are needed by the companies.

Demand for consumer durables

The demand for consumer durables upon

- a) changes in tastes and preferences
- b) fashion

c) demonstration effect: If one person is buying some product than one other would also want to buy that product ^{if even} if his income cannot ~~accommodate~~ ^{accommodate} that product.

d) cost of funds: interest rate

Consumer durables are the items which are not necessary for the consumers so the demand for these goods change with tastes and preferences whereas other goods which are necessarily consumed by the consumers that are consumer non-durables, the demand of these goods does not change much with tastes and preferences.

Investment in housing

A significant amount is invested in housing, $\therefore$ it affects the demand for funds.

Demand for consumer durables reflects both: the money demanded by the household sector and also the money demanded by the producers. Because when the demand for the goods will increase, household sector will demand more money to fulfill their demand and also the firms will expand their production for

which they will demand more funds.

* Financial Sector Development and Economic Growth

Two major channels of the financial sector:
Savings & investment.

- The Classical Prior Voluntary Saving Theory
- Credit Creation Theory
- Forced saving or Inflationary Financing Theory
- Financial Repression Theory
- Financial Liberalisation Theory

* Prior Savings Theory

fully Classical $\Rightarrow$ all the savings must be invested.

$$\text{Investment} = f(\text{savings})$$

Saving is a prerequisite or a determinant of investment.

It is averse to inflation, it advocates control of inflation. $\rightarrow$ Assumes 0 inflation in economy

If only the saved money is used for investment, then it will not have much effect on the money supply.

- growth is contributed through diff. transformation process
- Only the surplus money is circulated in the economy, and is utilized by the

Higher interest rate $\rightarrow$ high savings $\rightarrow$ high investments
(Real interest rate) = Interest rate - Inflation
(assumed to be low, @ 0)

production sector, $\therefore$ there are not much fluctuations in the money supply.

Suggests a policy of reasonably high positive real interest rates to encourage savings by the public.

As inflation is low, $\therefore$ even if interest rate is not much high, then also real interest rate ~~is~~ will be high. which will lead to higher savings & in turn higher investments.

Financial institutions promote development by offering the following "transformation services or functions" (ASMR)

1) Liability - Asset Transformation:

Banks convert ~~asset~~ savings (asset of people, liability of banks) to investment (asset of bank, liability of firms)

2) Size - Transformation

By pooling the small savings of many people, banks are able to give larger loans to firms which would have not been possible if funds were invested individually.

Risk Management ways

① Transfer risk ② Bear risk

3) Risk - transformation ③ Hedge risk

(speculate)

Banks and other financial institutions are able to diversify the portfolio. The

If portfolio is diversified,
risk is decreased

↑

↳ banks help do this

pooled money in the form of savings is later utilized for various purposes. If portfolio is diversified, risk is decreased, loans are given to many sectors for various purposes, ∴ if one sector / individual defaults, there are many other entities to compensate the loss.

Diversification of portfolio leads to reduced risk.

⇒ short term savings could be converted to long term debt through financial sector

4) Maturity - Transformation

Banks and other financial institutions provide the long-term loans against the short-term deposits. Deposits with the bank are generally short-term but ~~to~~ banks lend money for long-term, this is possible ~~is~~ for the bank because of the continuous flow of money.

Even when banks borrow from RBI or some other sources, the borrowing is for short term.

* Credit Creation Theory

Financial system plays a positive ~~role~~ and catalytic role by providing finance or credit through creation of credit in anticipation of savings.

This theory advocates for financial ~~sy~~

CRR $\rightarrow$ Cash Reserve Ratio $\rightarrow$ money deposited by bank in reserve bank of India

Initial deposit = 100

Cash Reserve = $10\% = 10$

Available for credit = 90 $\rightarrow$ Taken as loan

institutions to create ~~no~~ credit to provide loans for investment considering that savings will eventually come.

Independence of investment from saving in a given time period.

10/8/23

3- Credit creation theory

CRR $\rightarrow$ cash reserve ratio $\rightarrow$ money deposited by bank in reserve bank of India

Initial deposit = ₹100

Cash reserve = 10% (₹10)

Available for loan = ₹100 - ₹10 = ₹90 $\rightarrow$ Taken as loan
↓
Kept deposit

Again used for deposit

10% CRR

(₹9)

₹81 left for deposit

Credit multiplier = $\frac{1}{\text{CRR}}$ here $\frac{1}{0.1} = 10$

If initial money was ₹100, then total money in circulation will go till ₹1000. (100×10)

Economies of scale $\rightarrow$

Initial amount of ₹100 became ₹1000 in the system

This theory assumes

Savings = Investment

Equality in savings and investments

Investment out of created credit results in a prompt income generation.

★ Theory of Forced Savings $\rightarrow$ calls for inflation induced growth

~~Some~~ Some significant inflation is necessary for the investment to rise and the economy to grow.

2- Forced Savings Theory (Keynesian Theory)

- Savings is not a pre-requisite for investments
- Indogenous → supply is exogenous

Money supply is exogenous
← Exogenous → something that can be determined independently

as per Keynesian, regulatory authority could control the money supply on their own

↓
If money supply is increased, growth can automatically take place but will also lead to inflation

↓
Hence Keynes theories call for inflation induced growth

↓
If there is underemployment (unemployment) by inc ↑ money supply, those resources can be utilised

↓
If inflation goes up with full employment following will happen

1. Tobin's portfolio shift effect $\rightarrow$ shift from financial sector to real sector
 $\rightarrow$ income coming from financial sector to physical sector
 2. Income distribution effect $\rightarrow$ Income of one financial sector will flow to other sector (real)
 $\hookrightarrow$ profit $\uparrow$, savings $\uparrow$, money come back again to financial sector for investment
 3. Inflation tax effect $\rightarrow$ profit $\uparrow$, tax $\uparrow$, gov. can use more money on public spending
- $\rightarrow$ Due to $\uparrow$ inflation $\rightarrow$ people won't be willing to invest in financial sector due to low real returns

This will increase consumption, hence income of producers will $\uparrow$

$MS \uparrow$ Investment $\uparrow$ Output $\uparrow$ Income $\uparrow$ Savings $\uparrow$ $\rightarrow$ if resources are under-employed

Portfolio shift effect $\rightarrow$ if there is full employment

$\downarrow$

Still $MS \uparrow$

$\downarrow$

inflation $\uparrow$, real interest rate $\downarrow$

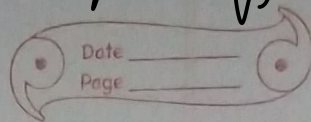
$\Downarrow$

people will shift to physical goods (in anticipation of further inflation)

$\downarrow$

consumption $\uparrow$, output $\uparrow$, savings $\uparrow$

Investment $\rightarrow$ can be increased autonomously (independently) through monetary expansion



$\rightarrow$ Savings is not a prerequisite for investments

Investment is not determined by savings.

Investment can be increased autonomously through monetary expansion.

$\hookrightarrow$ Money supply $\uparrow$ Investment $\uparrow$, output $\uparrow$, income $\uparrow$, savings $\uparrow$

If there are some unemployed resources in the economy and the money supply is increased then production will increase and aggregate demand will increase, income will increase and eventually savings will increase.

Tobin's Portfolio shift effect (if resources are fully employed)

Increase in money supply will lead to inflation $\rightarrow$ lower the real rate of return of on financial investments and money Real balances will go ^{up} ~~down~~ because people will not be interested to invest in the financial markets.

Instead, demand for physical capital i.e. consumer durables will rise and ~~prod~~ production of those goods will rise which will lead to increase in income and savings. $\therefore$ There is a shift of portfolio from financial to consumer durables market.

$\Rightarrow$ Tobin's portfolio shift effect.

$\hookrightarrow$ Shift from financial sector to physical goods / real sector

↳ If there is full employment $\rightarrow$ Still $MS \uparrow \rightarrow$ Inflation $\uparrow$
consumption $\leftarrow$ people will shift to physical goods $\leftarrow$ real interest rate $\downarrow$
output $\uparrow$
savings $\uparrow$ (in anticipation of further inflation)

Income distribution effect (increasing savings through profit)
 $\rightarrow$ Savings of profit-earners will rise

Inflation tax effect $\rightarrow$ profit $\uparrow$ Tax $\uparrow \rightarrow$ more money spent on public
Inflation leads to more money getting transferred to govt as tax which will finally go to more investments done by the govt.

★ Financial Regulation Theory

Financial markets are prone to market failure.

Certain forms of Government intervention are required.

The lowering of interest rates through government intervention improves the average quality of the pool of loan applications, and improves the efficiency with which capital is allocated.

Lower interest rates makes the loans cheaper for the firms and the total output will rise.

Financial Regulation Theory

There is always a chance, a market can lead to a failure

- All the markets are prone to failure
↓ what is failure of market?
no equality b/w supply & demand

who can resolve market failure?

→ Someone has to intervene to stabilize ⇒ government

Regulatory body

Banks → RBI

Insurance → IRDA (Insurance regulatory & development authority)

Mutual funds & stock

→ SEBI (Securities & exchange board of India)

→ PFRDA (Pension fund regulatory and development authority)

Assumes → if regulatory body → low interest rate

more growth

↓
more money taken for investment

Priority sector lending

↳ 40% of total money has to be lent to priority sector (agriculture, small scale industries)

Banks would be reluctant to provide loan to priority sector & so they are bound by regulatory body to ensure uniform growth.

Stable payment system

Payment systems become stable due to regulatory norms

- If 3rd party like regulatory body is there in a transaction it leads to more confidence among market participants
Banks have to pay fines & penalties

↳ to RBI incase of not complying

Every product becomes public good

ex yield curve

↳ graph of yield to maturity

so RBI said yield curve should be public
Facilities & services are available equally

Financial Liberalisation Theory

- Market should be completely open
- Demand & supply should decide price of product
- If liberalisation $\rightarrow$ then interest rates would be higher
 - $\downarrow$ why??
 - $\downarrow$ But not good for domestic investors
 - increase the interest rate $\rightarrow$ attract more foreign investors

ex US interest low $\rightarrow$ Investor borrow
India " high $\rightarrow$ Investor puts } he comes

$\downarrow$
Our investor suffers

Market interest rate will be higher $\rightarrow$ more competitiveness
due to liberalisation

even if price is higher $\rightarrow$ due to competitiveness

price will be

Due to liberalisation adjusted

- $\rightarrow$ Productivity of investment increases $\rightarrow$ due to competition
- $\rightarrow$ Allocation efficiency increases
- allocative